VIVEK KUMAR GARG

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Professional Experience

Data Engineering Intern – Offer Letter **May 2026 – Present**

moneyHop, Bangalore | Remote

- Developed **3 internal data platforms**, including an **AI-powered Enterprise SQL Assistant**, a **Manual and Automated Data Lineage Platform**, and an **Analytics Reporting System**, enabling **self-service data querying**, **data governance**, and **operational reporting**.
- **Built** scalable backend services using **Python, FastAPI, MCP (Model Context Protocol), Gemini API, MySQL, PostgreSQL, SQLAlchemy, JDBC, React, WebSockets, Docker, Google Sheets API, and AWS Lightsail**, implementing **schema discovery**, **metadata management**, **lineage automation**, **data profiling**, **AI-assisted SQL generation** that **reduced query generation time** by **40%**, **REST APIs**, and **cloud deployment**.

Software Developer Intern - Completion Certificate **October 2025 – March 2026**

Cardio Flow Design, Japan | Remote

- Delivered a **medical imaging desktop application** using **C#, WPF, .NET, and fo-dicom,** featuring DICOM parsing, **real-time image rendering**, cine playback for **500 MB+** high-resolution DICOM datasets, diagnostic measurement and annotation tools, **image transformations**, and report export using a **modular service-oriented architecture**. | GitHub | Demo Video

Founder - Patent and Pre-Incubation Letter **November 2024 – April 2026**

IndoMate, Jaipur | On-site

- Founded a **patent-backed**, **pre-incubated emergency response platform** featuring **5+ SOS modules for real-time user safety** and **incident reporting**. | Product Brief
- Presented the product at **7+ national-level innovation competitions**, including the **AICTE National Innovation Contest**, and secured **2nd Runner-up** at **E-Summit '25 Ideathon** while ranking **9th among 600+ teams** at **Startup Conclave**.

Software Engineering Intern - Offer Letter **May 2024 – May 2026**

Software Development Center – MUJ, Jaipur | On-site

- Engineered a **full-stack Elective Management System** serving **10,000+ students**, enabling **subject preference management**, **algorithmic elective allotment**, and academic administration through a centralized platform | Live Project | GitHub
- Developed scalable backend APIs and responsive frontend modules using **React.js, TypeScript, Node.js, Express.js, Prisma ORM, PostgreSQL, Docker, and CI/CD**, implementing **RBAC**, authentication, and **CI/CD deployment workflows**.

Freelance Full-Stack Developer

- Delivered **3+ full-stack web applications**, including custom **e-commerce platforms** and **business solutions**, managing the complete **development lifecycle** from requirements gathering to deployment.
- Built responsive, database-driven applications with **secure authentication**, **payment integration**, **admin dashboards**, **order management**, and REST APIs that processed **over 500 successful transactions,** using modern full-stack technologies | Live | GitHub

Programs & Open-Source Contributions - View Certificates

- **Amazon ML Summer School 2026**
- **HPE Career Preview Program (CPP3) 2026**
- **Deloitte Capstone Program 2025-26 (AI Career Guide Project)**
- **GirlScript Summer of Code (GSSoC) — Mentor (2025) | Contributor (2024)**

Technical Projects

Thread.ai – Real-Time Multimodal AI Interaction Platform | Demo Video | Live Project | GitHub

- Engineered a **real-time AI conversation system supporting video, voice, vision,** and context-aware understanding.
- Designed **3 interaction modes**: **Live AI Video Chat**, **Company Intelligence** (**RAG** powered by Pinecone), and **vision-based** contextual assistance. **Tech** Stack: React, Firebase, Pinecone (Vector DB), TensorFlow, Tesseract, Gooey AI
- Recognized as a finalist across **multiple national and international hackathons**, including **Top-9 Global Finalist at nextGenHacks**, **Finalist at India Innovates**, and **Product Stage qualifier at the Amazon AI for Bharat Hackathon**.

EffortX – AI-Powered Developer Reputation Verification Platform | Video | Live Project | GitHub

- Constructed an AI-driven system that **parses raw GitHub** diffs via **dual-auth (OAuth & GitHub App)** to **measure meaningful engineering impact rather than shallow activity metrics**.
- Implemented an **automated pipeline that generates context-aware markdown reviews** using **Gemini 2.5**, commits **immutable proofs onto the Solana blockchain**, and **unlocks premium multimodal AI tools** (via **Dappier AI**) using a custom token economy (ECOIN). **Tech Stack:** React, Node.js, Express.js, Redis, Solana (Anchor), Gemini AI, Dappier AI.
- Achieved a **top score of 88/100** in **the OutSkill x OpenAI Hackathon** and successfully submitted the platform to the **Solana Frontier Hackathon**. Development Journey

Stock Analyzer App – Interactive Virtual Trading & Financial Analytics | Video | Live Project | GitHub

- Architected an AI-driven trading education platform, **integrated 9+ software** tools, including **payment gateway**.
- Integrated multimodal assistance by embedding **conversational AIs** configured to **answer user financial queries** in natural language in the form of **text, AI-supported voice calls, and real-time AI-supported video calls**.
- **Tech Stack**: React, **RevenueCat, Tavus, ElevenLabs, Stripe**, **Lingo**, Yahoo Finance API, Finnhub API, Dappier.

Slangyy – Style-Adaptive AI Text Rewriter | Video | Live Space | GitHub

- Designed **a Chrome Extension** (Manifest V3) featuring a floating UI that **detects text selection** and dynamically communicates with a **background rewording API**.
- **Fine-tuned a Microsoft Phi-3 model** using Unsloth, **QLoRA**, and PEFT on a **Google Colab** T4 GPU, **reducing model size** to approximately **50 MB**, **deploying the quantized model** to a **Hugging Face Gradio Space**.
- **Tech Stack:** Chrome Extension API, **Hugging Face** (Spaces & **Transformers**), **Unsloth**, **PEFT**, **TRL**, Gradio, Python.

Technical Skills

- **Programming:** JavaScript, TypeScript, Java, C#, C, Python, SQL
- **Frontend:** React.js, Redux.js, Tailwind CSS, Chrome Extension API, Bootstrap, HTML5/CSS3, XAML, **Electron.js**, WPF, SCSS/SASS, EJS, Handlebars.js
- **Backend:** Node.js, Express.js, Django, Flask, FastAPI, WebSockets, SQLAlchemy, Prisma ORM, JDBC, REST APIs
- **Databases**: PostgreSQL, MySQL, MongoDB, Redis, Firebase, Supabase
- **AI/ML:** GenAI, MCP (Model Context Protocol), LangChain, LangGraph, RAG Systems, LLM APIs, Pinecone, Vector Databases, Prompt Engineering, Google AI Studio, OpenCV, TensorFlow, Tesseract, NLP, Unsloth, QLoRA, PEFT, TRL
- **Data Engineering:** Schema Discovery, Data Lineage, Metadata Management, Data Profiling
- **Cloud & DevOps:** AWS Lightsail, **Version Control,** Docker, Docker Compose, Kubernetes, Google Cloud IAM
- **Blockchain:** Solidity, Rust, Solana, Anchor Framework, Web3.js, Ethers.js, Clarinet, Smart Contracts
- **Tools & Platforms:** Git, GitHub, Postman, Figma, Hugging Face, VS Code, Vercel, Netlify, Antigravity

Education

Manipal University, Jaipur — **Expected May 2027**

B.Tech in Computer Science and Engineering - Transcripts — **CGPA: 9.28**

- **Honors: Dean's List (Recognized for 5 consecutive semesters)**
- **Awards: Recipient of the Student Excellence Award (2x)**
- **GATE CS 2026 Qualified, Score 577, All India Rank (AIR) – 3839**

Relevant Coursework

Core Computing & Software Development: Data Structures & Algorithms (DSA), Design & Analysis of Algorithms (DAA), Problem Solving using Computers, Object-Oriented Programming (Java), Operating Systems, Database Management Systems (DBMS), Computer Networks, Software Engineering.

Artificial Intelligence & Data Science: Machine Learning, Natural Language Processing (NLP), Computer Vision (CV), Predictive Analytics, AI & Soft Computing, Information Retrieval, Recommender System, Computational Mathematics, Statistics & Probability, Machine Learning with Python.

Systems, Security & Design: Cryptography, Computer Organization & Architecture, Automata & Compiler Design, Information Storage & Data Protection, UI/UX Design, Emerging Tools & Technology.

Awards & Achievements

Top-9 Global Finalist	**NextGenHacks International Hackathon**	**2026**
Global Finalist	**USAII Global AI Hackathon**	**2026**
Winner (1st Place)	**Technovate Project Expo (Ranked #1 AI Project across all tracks)**	**2026**
Winner (1st Place)	**UI/UX CyberSpace Hackathon (Live Link)**	**2026**
Winner (1st Place)	**404 Fund Not Found (TechFest MUJ Ideation Track)**	**2026**
Top-10 Finalist	**AMD Slingshot AI Innovation Challenge**	**2026**
National Semi-Finalist	**Flipkart Grid 8.0 (AI Engineering Track)**	**2026**
National Semi-Finalist	**ET-AI Hackathon (The Economic Times)**	**2026**
Winner (1st Place)	**MujHackX 3.0 Hackathon (Out of 500+ Participants)**	**2025**
National Finalist	**DigiGreen National Hackathon (4th Place out of 200+ teams)**	**2025**
Published Patent	**India Patent Office (No: 202411079707 — SOS System & Features)**	**2024**
Winner (3rd Place)	**BrandIT 2024 UI/UX Design Showdown**	**2024**

PROFESSIONAL CERTIFICATIONS

Generative AI: Google AI Essentials (Google), Fundamentals of Agentic AI (**Microsoft Learn**)

Frameworks & Core: React & Redux Mastery, Node.js, Express.js & MongoDB Suite (Scaler)

Full-Stack & Dev: Full-Stack Engineering Bootcamp (RiseIn), **Postman API Fundamentals Student Expert**

Core CS & DB: Data Structures & Algorithms (DSA) and Design & Analysis of Algorithms (DAA) (**Top 5% Rank — NPTEL**), OOP Using Java (**GeeksforGeeks**)

Assessments: SQL & PL/SQL Database Design (**Oracle**), Verified C# and DSA Specialist (**TestDome**)